Kaicheng Xue

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RESEARCH INTERESTS

Understanding and evaluating the behavior of large language models, from their internal mechanisms to their use as agents in the real world.

EDUCATION

University of Virginia, Charlottesville, VA Aug 2025 – Dec 2026 (expected)

M.S. in Computer Science, GPA: 4.0/4.0

Evaluating Entropy and Spectral Gap of Hidden-State Markov Chains as Predictors of LLM Reasoning Accuracy

Thesis advisor: Yangfeng Ji

Relevant Coursework: Machine Learning, Natural Language Processing, Trustworthy AI, Probabilistic Machine Learning, Data Privacy, Neural Networks, Geometry of Data

University of Wisconsin–Madison, Madison, WI Sept 2021 – May 2025

B.S. in Mathematics and Computer Science (double major), GPA: 3.79/4.0 (Mathematics 3.81, CS 3.78)

Relevant Coursework: Introduction to Algorithms, Introduction to Artificial Intelligence, Real Analysis, Stochastic Processes, Probability, Abstract Algebra, Linear Optimization, Linear Algebra, Discrete Mathematics

Honors: College of Letters & Science Dean's List

PUBLICATIONS

Xue, K., Zhang, X., Mao, R., & Cambria, E. "Understanding the Hidden State of Language Models from the Perspective of Sentiment Analysis." In *Proceedings of the 2024 IEEE International Conference on Data Mining Workshops (ICDMW)*, 2024. *(Workshop paper)*

RESEARCH EXPERIENCE

Graduate Research, University of Virginia Jan 2026 – Nov 2026

Evaluating Entropy and Spectral Gap of Hidden-State Markov Chains as Predictors of LLM Reasoning Accuracy

Advisor: Yangfeng Ji; manuscript in preparation

- Modeled hidden-state trajectories of nine Llama and Qwen3 models over reasoning steps as Markov chains, using entropy rate and spectral gap as accuracy predictors
- Evaluated the predictors on four reasoning benchmarks with nested cross-validation and permutation tests, including LoRA-fine-tuned variants
- Found that model size and family explain most accuracy variation on two of four datasets, and that, in a check on Llama-8B for HotpotQA, a step's state was almost fully determined by its position

Student Researcher, University of Virginia Aug 2026 – Present

LLM Abuse in Digital Advertising Marketplaces: Detection and Mitigation

PI: Tianhao Wang (CS); Co-PIs: Michael Albert (Darden), Jingjing Li (McIntire)

- Build a simulated Amazon-style marketplace populated by LLM shopper agents, with personas matched to real Amazon buyers
- Implement an LLM-generated fake-review attack that combines push, segment-targeted, and average-profile reviews, and compare shopper behavior with and without it
- Building on real shopper data, explore how to make shopper agents behave more like humans

Course Research Project, University of Virginia Spring 2026

Privacy Attack via Temporal Embedding Inversion

Course: Data Privacy; Instructor: Tianhao Wang

- Proposed a sliding-window method that uses dialogue history to invert conversation embeddings leaked from RAG systems
- Raised attack success over the baseline with a fine-tuned local embedder, e.g., from 15% to 41% on user profiles, on a dataset of profile, financial, and medical conversations
- Analyzed why a non-linear extension fails to beat the linear method, attributing it to distorted embedding-space geometry

Research Intern, Nanyang Technological University May 2024 – Sept 2024

Understanding the Hidden State of Language Models from the Perspective of Sentiment Analysis

Supervisors: Xulang Zhang, Rui Mao, Erik Cambria

- Analyzed hidden states of a fine-tuned RoBERTa across layers and the attention heads of layer 11 with K-means, spectral, and hierarchical clustering
- Found that clustering separates sentiment mainly in the last three layers (about 80% accuracy at layer 12 versus about 50% earlier)
- Showed that early-to-mid layers encode part-of-speech information, and that across word sense disambiguation and entailment the mechanisms are task-driven rather than human-like

TECHNICAL SKILLS

Languages: Python, Java, C, SQL, JavaScript
Frameworks & Tools: PyTorch, Hugging Face Transformers, scikit-learn, NumPy, pandas, MATLAB
Infrastructure: Git, Linux, CUDA, Docker, AWS, LaTeX
Methods: Hidden-state probing, Markov chain analysis, stochastic processes, spectral analysis, clustering (K-means, spectral, hierarchical), benchmark construction, statistical evaluation (permutation tests, cross-validation)